Van Ginneken

Van Ginneken is the classic DP for buffer insertion on trees

The idea

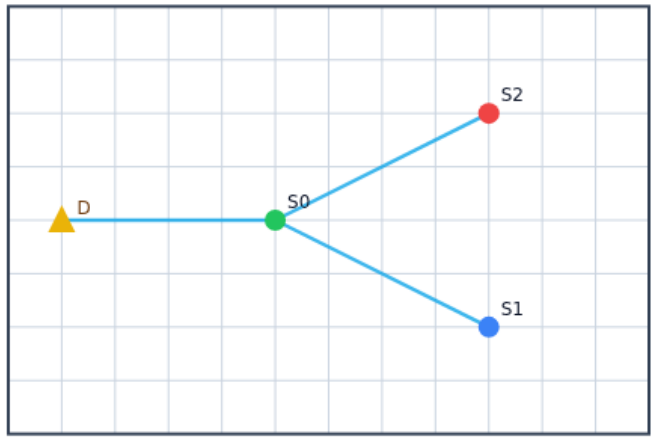
- For each edge build wire-only and mid-buffer options as (c,d) pairs
- Dominate_prune keeps non-dominated fronts
- Pick minimum delay
- Sum along the toy tree

Options

VAN GINNEKEN

Options

Step 1 / 5 · opts · module02-05-van-ginneken



Explanation

Wire-only plus one option per library buffer at mid.

- (c,d) pairs

```
DP lite
```

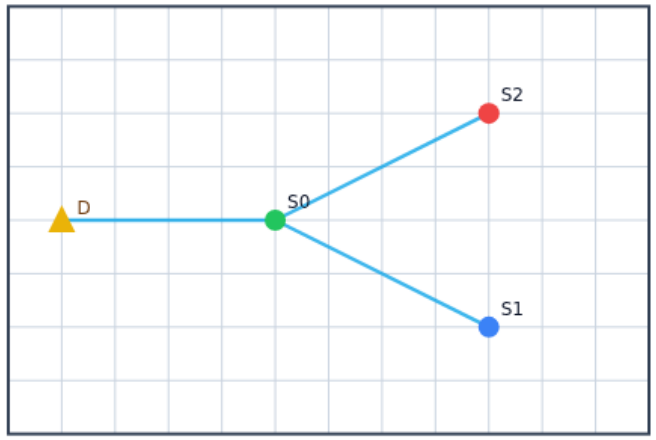
Options

Prune

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Prune

Step 2 / 5 · prune · module02-05-van-ginneken



Explanation

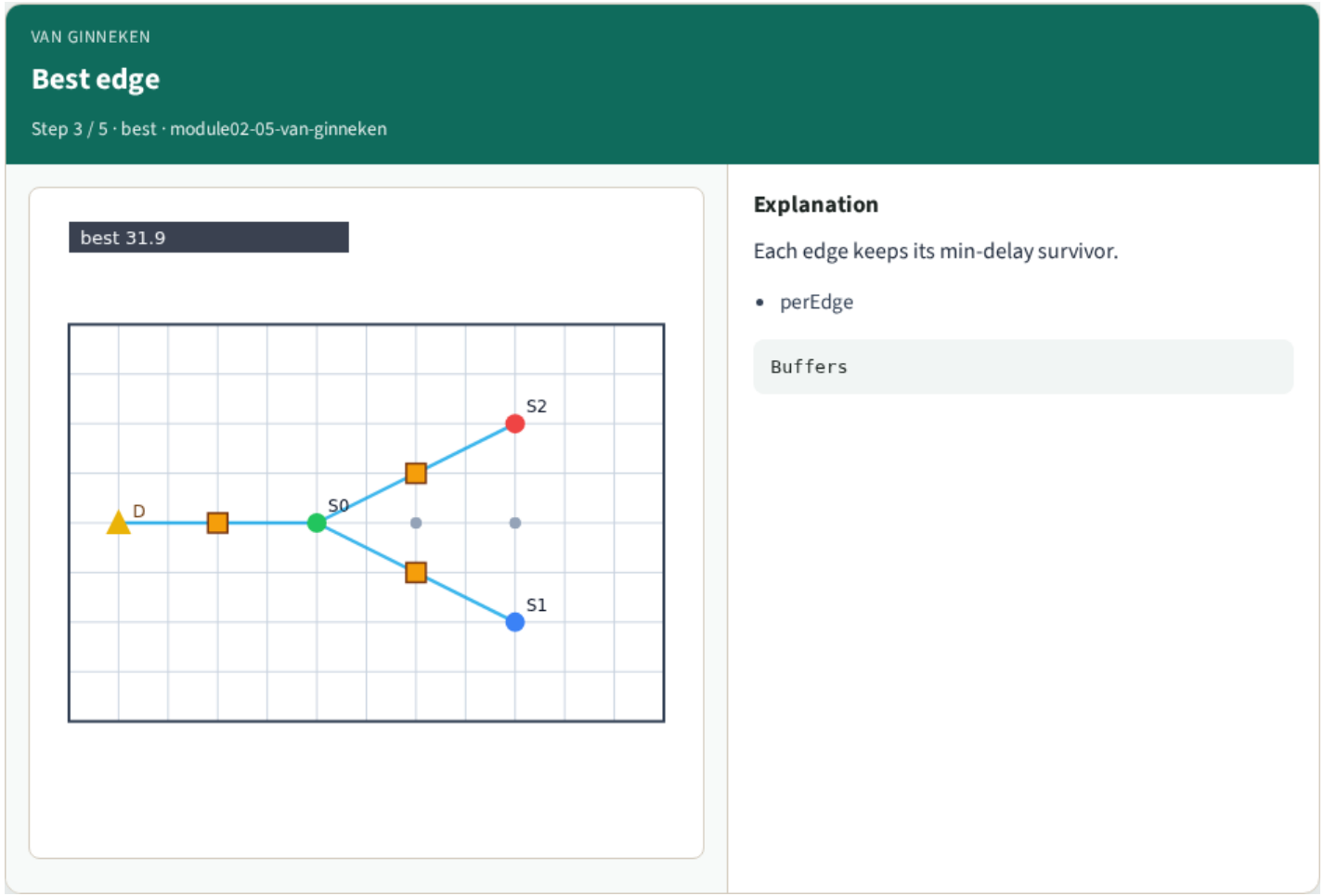
Drop pairs with worse cap and worse delay.

- dominatePrune

```
Pareto front
```

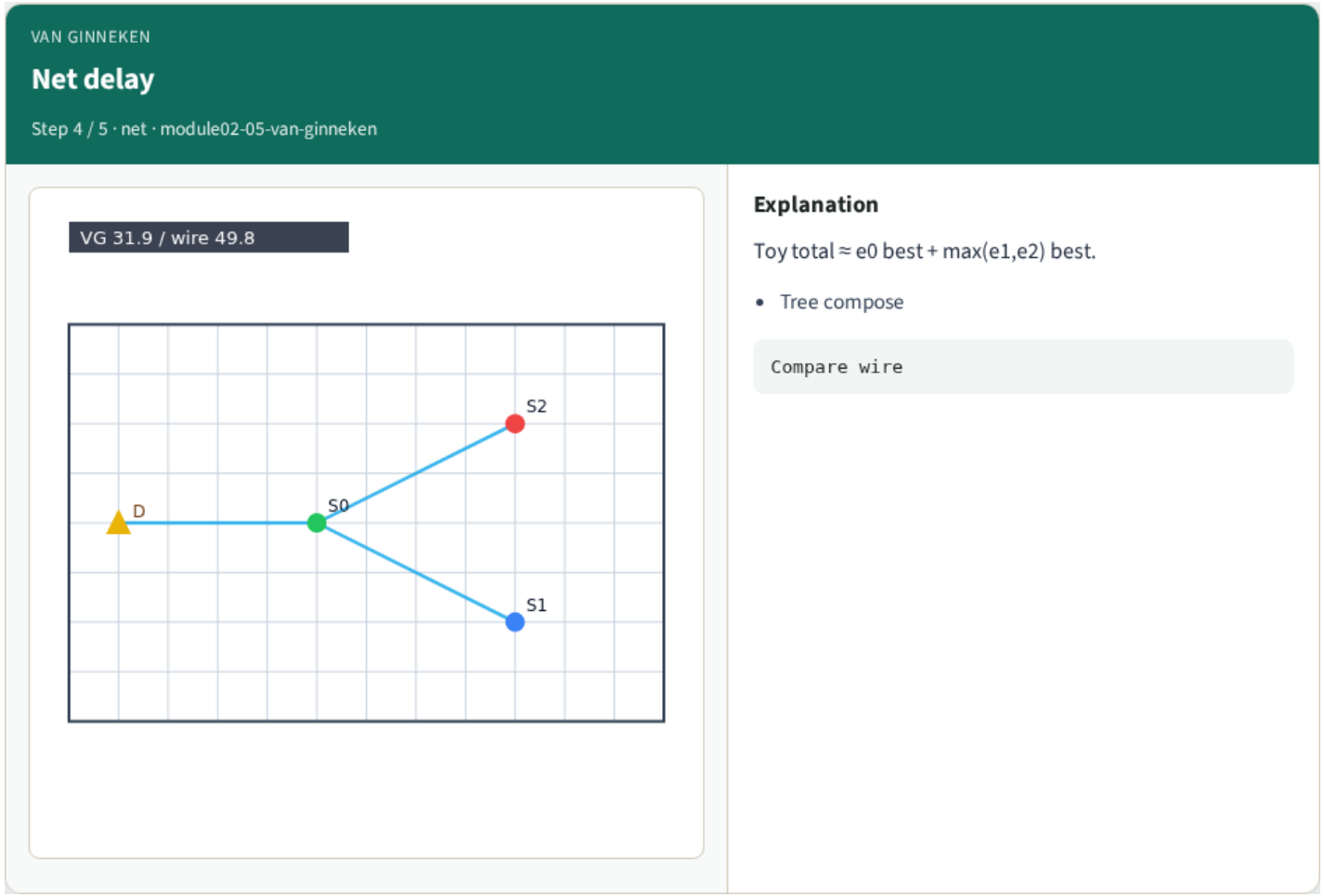
Prune

Best edge



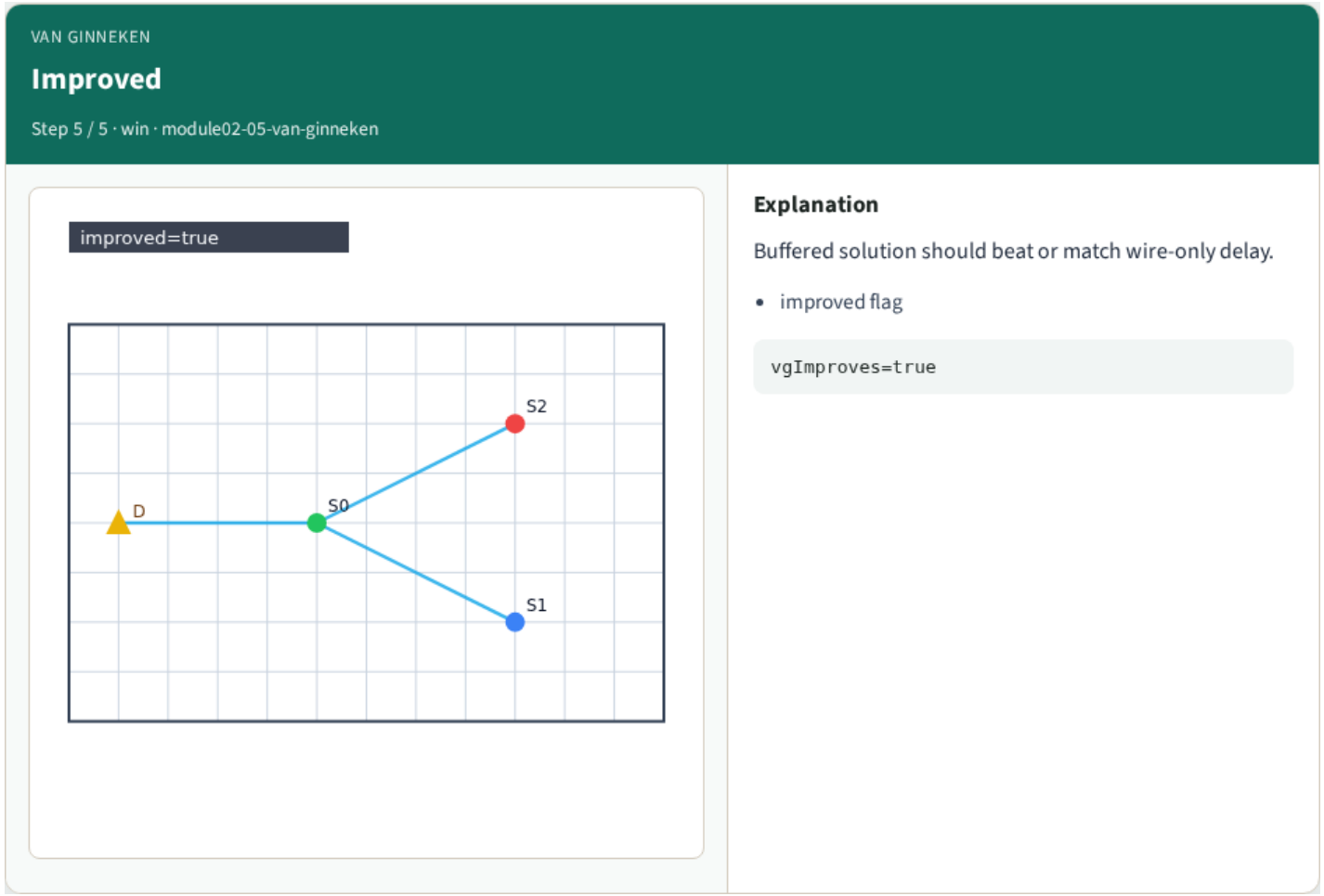
Best edge

Net delay



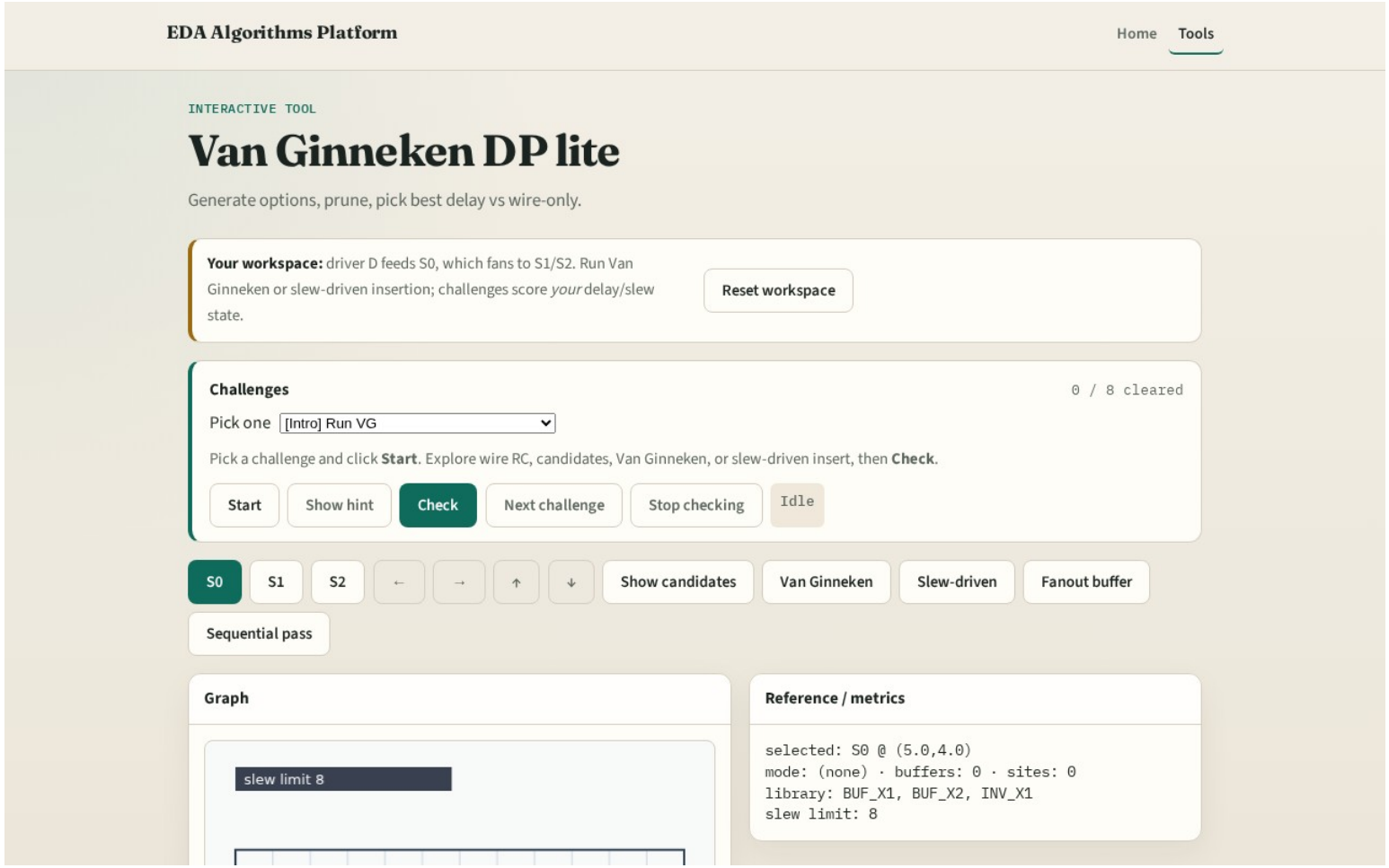
Net delay

Improved



Improved

Browser lab track



Browser lab starter

Implement track

- Call `van_ginneken_net(data)`
- Assert best_delay ≤ wire_delay

Pitfalls

- Keeping dominated options
- Double-counting branch loads
- Forgetting wire-only option

Your turn

- Next: slew-driven greedy insertion